a bit clearer

C PROGRAMMING FROM SCRATCH · EPISODE 08

Data Types

char, int, float and double

KEY POINTS

- A type tells the compiler what kind of value it is working with.
- char stores characters written in single quotes; int stores whole numbers.
- float and double store decimals; double usually has a wider range and more significant digits.
- A plain decimal like 3.14 is a double; 3.14f is a float.

01 Data comes in kinds

▶ VIDEO 00:43

Sort a receipt by kind: the quantity `3` is a whole number, the price `5.50` and subtotal `16.50` are decimals, and the product name is text. The compiler can only store and work with a value once it knows its type.

02 The map of C types

▶ VIDEO 01:32

CHARACTER	INTEGER	FLOATING-POINT	BOOLEAN
char	**int**	**float, double**	**true / false**

From the built-in types, C builds more: pointers, arrays, structures, unions and enumerations.

03 char and int

▶ VIDEO 01:59

`char` comes from **char**acter, `int` from **int**eger. A single character goes between single quotes.

Excerpts: char-letter.c, int-num.c — C SOURCE

```c
char letter = 'A';
printf("%c\n", letter);   /* A  */

int num = 35;
printf("%d\n", num);      /* 35 */
```

04 Why "floating point"?

▶ VIDEO 03:02

The decimal point can float: **123.45 = 12.345 × 10 = 1.2345 × 10²**. In C, `e2` means × 10².

floating-point.c C SOURCE

```c
#include <stdio.h>

int main(void)
{
    double a = 123.45;
    double b = 1.2345e2;
    printf("%f\n", a);
    printf("%f\n", b);
    return 0;
}
```

05 float and double

▶ VIDEO 03:48

TYPE	PRECISION	LITERAL
`float`	single	`3.14f`
`double`	double	`3.14`

> KEY IDEA **A plain decimal is a double**
>
> Write `3.14` and the compiler treats it as a `double`. Add `f` to make it a `float`.

06 All three together

▶ VIDEO 04:26

three-types.c C SOURCE

```c
#include <stdio.h>

int main(void)
{
    char grade = 'A';
    int quantity = 3;
    double price = 5.5;

    printf("%c\n", grade);
    printf("%d\n", quantity);
    printf("%f\n", price);
    return 0;
}
```

`%f` prints six digits after the decimal point, so `5.5` appears as `5.500000`.

CODE 5 files on GitHub github.com/abitclearer/c-programming-from-scratch/tree/main/08-data-types